

Exponents and Scientific Notation39 points · two class periods **ANSWER KEY**

Name

Date

Period

*Show your work. Circle your final answer.***Day 1**

Sections 1 and 2 — numerical bases · 19 points

1. Laws of Exponents

9 points

1. Write each expression as a single power.

a. $4^3 \cdot 4^5$ 1 pt
 4^8

Add the exponents; the base stays 4.

b. $(2^3)^4$ 1 pt
 2^{12}

Multiply the exponents.

c. $\frac{7^9}{7^4}$ 1 pt
 7^5

Subtract the exponents.

d. 9^0 (give its value) 1 pt
 1

*Any nonzero base to the zero power is 1.**Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.*

2. Find the value of each expression.

a. $\left(\frac{2}{3}\right)^3$ 1 pt
 $\frac{8}{27}$

b. $6^0 \cdot 5^2$ 1 pt
 25

1 · 25. A 0 here means 6^0 was read as 0.

c. $\frac{(10^2)^2}{10^3}$ 1 pt
 10

 *$10^4 \div 10^3 = 10^1$.**Scoring: a 1 · b 1 · c 1 = 3 points.*

3. Which expression is equivalent to $2^4 \cdot 2^3$?

1 pt

- ☒ A. 2^7
☐ B. 4^7
☐ C. 2^{12}
☐ D. 4^{12}

A

Scoring: 1 point.

4. Find the value of n that makes the statement true: $\frac{5^n}{5^2} = 125$

1 pt

$n = 5$

$125 = 5^3$, so $n - 2 = 3$.

Scoring: 1 point.

2. Integer Exponents

10 points

5. Rewrite each expression with positive exponents. Do not evaluate.

a. 6^{-3}

1 pt

$\frac{1}{6^3}$

b. $\frac{1}{(-2)^{-5}}$

1 pt

$(-2)^5$

The power moves up; the base keeps its sign and its parentheses.

c. $3 \cdot 10^{-4}$

1 pt

$\frac{3}{10^4}$

The 3 is a factor, not a base with an exponent; it stays up.

Scoring: a 1 · b 1 · c 1 = 3 points.

6. Find the value of each expression.

a. 4^{-2}

1 pt

$\frac{1}{16}$

Reciprocal of 4^2 . -16 is the misconception; -8 multiplied base by exponent.

b. $(-3)^{-3}$

1 pt

$-\frac{1}{27}$

1 over $(-3)^3 = 1$ over -27 .

c. $\left(\frac{1}{2}\right)^{-3}$

1 pt

8

Flip to 2, then cube.

d. $2^5 \cdot 2^{-7}$

1 pt

$2^{-2} = \frac{1}{4}$

$5 + (-7) = -2$.

Scoring: $a \cdot 1 \cdot b \cdot 1 \cdot c \cdot 1 \cdot d \cdot 1 = 4$ points.

7. Which is the value of 5^{-2} ?

1 pt

☒ A. $\frac{1}{25}$

☐ B. -25

☐ C. -10

☐ D. $\frac{1}{10}$

A

Scoring: 1 point.

8. Select ALL of the expressions whose value is $\frac{1}{64}$.

1 pt

☒ A. 2^{-6}

☒ B. 4^{-3}

☐ C. -8^2

☒ D. $\left(\frac{1}{4}\right)^3$

☐ E. 2^{-8}

A, B and D

All three correct and nothing else for the point.

Scoring: 1 point.

9. Marco says the value of 3^{-2} is -9 . Explain why he is wrong, and give the correct value. 1 pt

A negative exponent means the reciprocal of the power, not a negative number. $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$.

The point needs the correct value AND the reason (reciprocal, not opposite). 'He forgot parentheses' is about $(-3)^2$, a different expression.

Scoring: 1 point.

Day 2

Sections 3 and 4 — variable bases, and scientific notation · 20 points

3. Variable Bases

9 points

10. Write each expression as a single monomial.

a. $x^4 \cdot x^6$ 1 pt

x^{10}

b. $(y^3)^5$ 1 pt

y^{15}

c. $(3a^2)^3$ 1 pt

$27a^6$

The 3 is cubed as well. $3a^6$ and $9a^6$ are the two coefficient errors.

d. $\frac{15m^7}{5m^2}$ 1 pt

$3m^5$

Coefficients divide; exponents subtract.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

11. Rewrite each expression using only positive exponents.

a. b^{-4} 1 pt

$\frac{1}{b^4}$

b. $5x^{-2}$ 1 pt

$\frac{5}{x^2}$

Only the x moves; $1/(5x^2)$ moved the coefficient too.

c. $(2m^{-3}n)^2$ 1 pt

$\frac{4n^2}{m^6}$

$4m^6n^2$, then m^6 moves down.

Scoring: a 1 · b 1 · c 1 = 3 points.

12. Which expression is equivalent to $(4x^3)^2$? 1 pt

☒ A. $16x^6$

☐ B. $8x^6$

☐ C. $16x^5$

☐ D. $4x^6$

A

Scoring: 1 point.

13. How are x^2 and x^{-2} related? Explain, and give the value of each when $x = 5$. 1 pt

They are reciprocals: $x^{-2} = \frac{1}{x^2}$. **When $x = 5$, $x^2 = 25$ and $x^{-2} = \frac{1}{25}$.**

The point needs the word reciprocal (or 1 over x^2) and both values. -25 for x^{-2} is the misconception.

Scoring: 1 point.

4. Scientific Notation

11 points

14. Write each number in scientific notation.

- a. 8,300,000 1 pt

8.3×10^6

- b. 0.00047 1 pt

4.7×10^{-4}

The 4 is in the fourth decimal place; 10^{-3} counted the zeros.

- c. 60,500,000,000 1 pt

6.05×10^{10}

The interior zero stays in the coefficient.

- d. 0.0000000092 1 pt

9.2×10^{-9}

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

15. Write each number in standard form.

- a. 2.9×10^5 1 pt

290,000

- b. 7.1×10^{-6} 1 pt

0.0000071

Scoring: a 1 · b 1 = 2 points.

16. Which is 0.0036 written in scientific notation? 1 pt

- ☒ A. 3.6×10^{-3}
☐ B. 3.6×10^{-2}
☐ C. 36×10^{-4}
☐ D. 3.6×10^3

A

Scoring: 1 point.

17. Answer each question.

- a. How many times larger is 9×10^9 than 3×10^6 ?

1 pt

$3 \times 10^3 = 3,000$ times

$9 \div 3 = 3$ and $10^9 \div 10^6 = 10^3$. '3 times' ignored the powers; '30' read 10^3 as 10.

- b. How many times larger is 4.8×10^{-3} than 1.6×10^{-7} ?

1 pt

$3 \times 10^4 = 30,000$ times

$4.8 \div 1.6 = 3$ and $-3 - (-7) = 4$.

Scoring: a 1 · b 1 = 2 points.

18. Which is greater, 8.9×10^{-4} or 2.1×10^{-3} ? Explain how you know.

1 pt

2.1×10^{-3} — the exponent -3 is greater than -4 , so its power of 10 is larger; $0.0021 > 0.00089$.

The point needs the choice and the reason. 8.9×10^{-4} compared the coefficients.

Scoring: 1 point.

19. A computer displays a number as 5.6E8. Write the number in standard form.

1 pt

560,000,000

E8 means $\times 10^8$.

Scoring: 1 point.

Score Tracker

39 points, one point per lettered part. Section 1 — 9 · Section 2 — 10 · Section 3 — 9 · Section 4 — 11

Benchmark	Questions	Possible	Earned
MA.7.NSO.1.1	1, 2, 3, 4	9	
MA.8.NSO.1.3	5, 6, 7, 8, 9	10	
MA.8.AR.1.1	10, 11, 12, 13	9	
MA.8.NSO.1.4	14, 15, 16, 17, 18, 19	11	
Total		39	

Follow-through. If a student writes a wrong power in one part and then evaluates their own power correctly in the next, the next part still earns its point. Mark each part against what they wrote, not against the key.